# Hardware Troubleshooting Guide

This guide covers hardware troubleshooting for laptops, desktops, monitors, printers, docking stations, and peripherals.

## 1. Laptop and Desktop Issues

### 1.1 Computer Won't Turn On

- Check the power cable is securely connected at both ends.
- For laptops: try a different power adapter and check if the charging LED is lit.
- Perform a hard reset: hold the power button for 10 seconds until it turns off, then press power again.
- Battery issue (removable laptop battery): remove battery, hold power for 30 seconds, reinsert battery.
- If power LED blinks but no display: connect an external HDMI monitor to test the display.
- If none of the above work: submit a hardware replacement ticket. SLA: 4 business hours.

### 1.2 Computer is Very Slow

- Check Task Manager (Ctrl+Shift+Esc) and look for processes using over 80% CPU or RAM.
- Common culprits: Windows Update running in background, antivirus scan, too many browser tabs.
- Restart the computer using a full shutdown (hold Shift while clicking Shut Down in Windows).
- Check disk space: C drive should have at least 15% free. Delete temp files using cleanmgr as Admin.
- SSD health check: Run in Admin CMD: wmic diskdrive get status. If not OK, report immediately.
- RAM check: Start menu, search Windows Memory Diagnostic, select Restart now and check for problems.

### 1.3 Blue Screen of Death (BSOD)

- Record the stop code displayed on the BSOD screen before the computer restarts.
- After restart, check Event Viewer: Run, eventvwr, Windows Logs, System, filter by Critical.
- Run SFC scan: open CMD as Admin and run: sfc /scannow
- Run DISM: DISM /Online /Cleanup-Image /RestoreHealth
- Submit a ticket with the stop code and Event Viewer logs. IT diagnoses within 2 hours.

### 1.4 Overheating

- Ensure ventilation vents are not blocked. Do not use laptops on beds or soft surfaces.
- Clean vents with compressed air every 6 months.
- Safe CPU temperature is below 85 degrees Celsius under load. Install HWMonitor to check.
- If CPU consistently above 90 degrees C, submit a hardware service ticket for thermal paste replacement.

## 2. Monitor and Display Issues

### 2.1 No Display / Black Screen

• Check cable connections: unplug and firmly reconnect DisplayPort, HDMI, or VGA cable at both ends.

• Try a different cable and a different port on the monitor.

• Check if the monitor is powered on. Power LED should be white or blue, not amber or off.

• Cycle monitor input source via the monitor OSD menu to verify correct input selection.

• Press Windows + P to cycle display modes: PC only, Duplicate, Extend, Second screen only.

### 2.2 Resolution and Scaling Issues

• Right-click desktop, Display Settings, Resolution, set to Recommended.

• For 4K monitors: set scaling to 150% or 175% for comfortable readability.

• If resolution options are missing: update graphics drivers via Device Manager, Display Adapters.

### 2.3 Multi-Monitor Setup

• Connect monitors and press Windows + P, select Extend.

• Arrange monitors: Display Settings, drag monitor boxes to match physical layout.

• Set primary monitor: right-click the monitor box in Display Settings, Make this my main display.

• Docking station supports up to 2 external monitors via DisplayPort and HDMI simultaneously.

## 3. Printers

### 3.1 Adding a Network Printer

• Go to Settings, Devices, Printers and Scanners, Add a printer or scanner.

• If printer not found: click The printer I want isn't listed, then Add by IP address.

• Printer IP addresses are listed at https://intranet.company.com/printers

• Use the PCL6 Universal Driver available from the IT portal.

### 3.2 Print Jobs Stuck in Queue

• Cancel all jobs: Settings, Printers and Scanners, Open print queue, Cancel all documents.

• Clear print spooler via Admin CMD:

```
net stop spooler del /Q /F /S "%systemroot%\System32\spool\PRINTERS\*.*" net start spooler
```

• Power cycle the printer (turn off for 10 seconds) and retry.

### 3.3 Poor Print Quality

• Run the printer built-in cleaning cycle via the printer control panel or web interface.

• Check toner and ink levels: access printer web UI at http://[printer-ip].

• Request toner replacement via: helpdesk.company.com, Request Supplies.

## 4. Docking Stations and Peripherals

### 4.1 Docking Station Not Recognized

• Disconnect and firmly reconnect the USB-C or Thunderbolt cable.

• Try a different USB-C port on the laptop.

• Update dock firmware from the manufacturer website (HP, Dell, or Lenovo based on your model).

• Supported docks: HP Thunderbolt G4, Dell WD22TB4, Lenovo ThinkPad USB-C Gen2.

### 4.2 Keyboard / Mouse Not Working

- Wired: unplug and replug USB cable, try a different USB port.
- Wireless: check batteries, re-pair using the device pairing button.
- Bluetooth: Settings, Bluetooth, Remove the device, then re-pair from scratch.

## 5. Hardware Replacement and Procurement

### 5.1 Requesting Hardware Replacement

- Portal: helpdesk.company.com, New Request, Hardware, Replacement.
- Include: device serial number (sticker on bottom of laptop), fault description, urgency level.
- SLA: Business-critical replacements 4 hours. Standard replacements 2 business days.

### 5.2 Approved Hardware Catalog

- Laptops: Dell Latitude 5540, HP EliteBook 840 G10, Lenovo ThinkPad X1 Carbon.
- Desktops: Dell OptiPlex 7010, HP EliteDesk 805.
- Monitors: Dell U2723D (27 inch 4K), LG 27UN880-B (USB-C).

INFO: Personal hardware cannot be connected to the corporate network without prior IT security approval.